Response to Reviewer FBhK

BioCoLoop, submission 5046 | Revision dated 9 November 2026. Prepared for author review; not posted to OpenReview.

We have added the requested same-access controls and validation diagnostics, and revised the claims to distinguish collaborative access from the value of research history.

1. Random and direct search with identical laboratory access

The new main-text search table compares fixed configuration, random search without replacement, GP-EI Bayesian optimization, and direct/history-guided Qwen and Luna search. Every method uses the same ten training laboratories, ten development panels, twelve-design library, initialization rule and checkpoint-selection rule. Searches have six proposal slots after D00 and a maximum of 700 aggregation rounds.

We verified all 60 task/configuration fits and sealed the development-only choices before scoring the additional controls. Exact integration over all 462 six-design subsets gives random search's expected score and policy variability. The study is retrospective and conditional on one training seed per endpoint (42 for DTI; 61 otherwise), not 462 independent training experiments.

The results distinguish the mechanisms clearly. On Norman, Luna history reaches 23.11 Top-1, versus 22.30 for direct Luna and GP-EI and 22.66 in expectation for random search. On DTI, GP-EI/direct Luna instead reach 94.60 AUROC, versus 94.27 for both history methods and 94.45 for random search. On VCC, all LLM rows and GP-EI reach 31.50, but the fixed configuration already reaches 31.50; proteomics is entirely tied. Thus these controls do not establish a general history advantage, and we no longer use the access-scenario table to imply one.

Appendix E also includes a shared-first-proposal random control and controls matching realized fit counts. This matters because Qwen's invalid proposals reduce actual training to 600 rounds for proteomic history search and 500 for both Tahoe searches. In the latter case, random search matched to four valid proposals still reaches 26.30 in expectation, versus Qwen's 25.37. Failure costs are preserved rather than replaced by additional valid proposals.

2. Development-test divergence

We reconstructed all 14 distinct checkpoints selected at the recorded VCC/Tahoe prefixes, re-evaluated 140 development panels, and independently rescored all 80 archived prefix test results. The original scores are reproduced.

Aggregation mismatch does not explain these cases. Each VCC development panel contains exactly two interventions and each Tahoe panel six, so equal-panel macro Top-1 is mathematically identical to pooled intervention-macro Top-1. The development sets nevertheless contain only 20 and 60 independent interventions. Qwen/VCC's 1.17-point development gain accompanies an 11.89-point test decrease; Tahoe's gain corresponds to three additional correct development compounds, while test Top-1 falls by 11.94 points.

We added a development-only sensitivity experiment with 20 newly fixed candidate-roster seeds. Each roster resamples four distractors within the same development pool and shuffles option order; initial and already selected final checkpoints receive identical rosters. No model is trained or reselected. Qwen/VCC's original +1.17-point advantage becomes -1.34 ± 7.37 points across rosters, positive in 6/20 draws; Tahoe's +5.00 becomes -1.33 ± 4.82, positive in 7/20. These are roster-draw standard deviations, not training-seed or biological replication uncertainty.

This identifies sensitivity to the fixed comparison sets, alongside limited development intervention coverage. It does not establish the separate causal contributions of repeated selection and intervention distribution shift. The new rosters reuse the original development interventions; robustness across independently regrouped train/development/test partitions remains unestablished.

3. Allocation controls and revised conclusions

Sections 4.3 and 4.7 now distinguish the access scenario from same-access search. Section 4.8 diagnoses the validation failure, and the efficiency discussion retains both Norman's favorable and Tahoe's unfavorable retrospective examples.

We also completed 360 fresh fits comparing uniform, score-only, learning-rate-stratified score-only, random and trajectory-based allocation, each at 800 rounds and 160 development evaluations. This reproduces the original 10.84-point Norman gain over uniform. However, score-only promotion reaches 21.70 and the exact random-promotion expectation is 22.63, versus 22.32 for the trajectory policy. On Tahoe the trajectory policy is better (22.89 versus 20.90 for score-only and 21.95 for random), while random is better on VCC. Section 4.4 and Appendix E.5 therefore attribute no universal advantage to the trajectory criterion. These results remain separate from LLM proposal history throughout the abstract and conclusions.

Response to Reviewer AfJB

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Thank you for identifying the relevant prior work and the need to separate model-development benefits from additional laboratory access. We have revised both the positioning and the experimental comparison rather than treating the original main table as evidence for the history mechanism.

1. Relation to existing collaborative research systems

The revised related-work section discusses all three systems explicitly. The Virtual Biotech coordinates specialist agents around biomedical evidence for therapeutic decisions. AutoScientists maintains shared experimental state, including unsuccessful directions, and coordinates subsequent experimentation. Helmsman is directly relevant to automated federated design: its planning, coding and simulation-based diagnosis already connect LLM agents to federated system construction and refinement. We have updated its citation to the ICLR 2026 publication.

BioCoLoop's contribution is the biological model-development interface and its controlled evaluation. Task adapters connect interaction, observed-response proteomic and perturbation predictors to the same local-fitting, development-evaluation and proposal interfaces. The coordinator distinguishes training-example-weighted parameter updates from equal-panel development evidence. Source-specific heads allow studies with different response objectives to contribute to a shared representation while the target development population remains explicit. The evaluation then separates laboratory participation, proposal-history conditioning and training allocation.

This is a narrower and more concrete contribution than inventing shared research memory or automated federated design. We now state the four-factor finite search space directly, and do not portray it as open-ended architecture or source-code discovery. The abstract, introduction and conclusion have been rewritten around the implemented framework and the evidence for each decision.

2. Does the history mechanism itself improve search?

We added random search without replacement and fixed GP-EI Bayesian optimization under the same ten-laboratory access as direct and history-guided LLM search. All methods use the same twelve configurations, development panels, initialization and six-proposal cap. The new main-text table reports all five endpoints; Appendix E provides exact random-search variability, shared-first-proposal and realized-fit-count controls.

The result is not a general history advantage. Qwen direct and history select equally scoring models on all five endpoints. Luna history improves Norman from 22.30 to 23.11 Top-1, versus a random expectation of 22.66, but decreases DTI from 94.60 to 94.27 AUROC. GP-EI matches direct Luna on DTI. Proteomics retains the starting configuration, and the VCC score attained by the LLM searches is already attained by the fixed configuration. These controls directly address the alternative explanation that conventional configuration search is sufficient in much of this small design space.

The comparison reuses verified complete candidate fits, with development-only selections sealed before additional scoring. It is a retrospective one-training-seed-per-endpoint study, not new multi-seed evidence. Three-seed source-defined laboratory controls are also reported separately: Qwen history is below direct on both proteomics and cell prediction, and the additional Luna feedback experiment yields only a small protein change and cell ties. We retain these results rather than extending the combined-system gain to unsupported component claims.

3. What the revised empirical contribution supports

The original main table is now explicitly an access-scenario comparison, with a Labs column. The new same-access table addresses search quality. Re-evaluation and new development-roster diagnostics identify validation sensitivities associated with poor held-out generalization.

We also completed 360 fresh fits to compare the allocation policy with uniform, ordinary score-only, learning-rate-stratified score-only and random promotion at identical training/evaluation counts. The original Norman gain over uniform is reproduced, but score-only promotion recovers most of it (21.70 versus 22.32), and the exact random-promotion expectation is 22.63. The trajectory policy improves Tahoe relative to both alternatives, while random is better on VCC. We report these mixed controls in Section 4.4 and Appendix E.5 rather than claiming a generally superior new allocation algorithm. The revised contribution is the implemented biological interface and its controlled empirical assessment, with data-access, search and allocation effects distinguished.

Response to Reviewer cR5g

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Thank you for the concrete concerns about conventional tuning, laboratory heterogeneity and the interpretation of the baseline comparison. We have added the missing controls and reorganized the evidence around those questions.

1. Conventional hyperparameter optimization

The revision adds a same-access table comparing random search without replacement, fixed GP-EI Bayesian optimization, direct LLM search and history-guided LLM search. All use ten laboratories, identical development panels, the same twelve-design library and a six-proposal cap after the initial configuration. Exact enumeration of 462 candidate subsets reports random search's expected outcome and search-policy variability; it does not count these subsets as independent training seeds.

For example, DTI AUROC is 94.60 for GP-EI, 94.45 in expectation for random search and 94.27 for history-guided search. On Norman, Luna history reaches 23.11 Top-1, versus 22.30 for GP-EI/direct and 22.66 for random search. Other endpoints include ties and cases where the fixed configuration is already competitive. Thus the new evidence supports task-dependent search behavior, not general superiority to conventional tuning. We also report controls matching the actual number of valid fits when an LLM proposal fails.

2. Laboratories defined by different studies

The main ten-client experiment partitions one source dataset; it is not ten independently operated institutions. We now make this explicit and separate it from the source-defined laboratory experiments.

The proteomic experiment treats PTPC, Lin, Ruprecht and decryptE as four laboratories. Their different response objectives use source-specific heads and a shared representation; the same target development panels select checkpoints. In the fixed-design transfer comparison, adding the three external studies increases target AP from 34.41 to 37.53.

The cell control compares four same-source VCC groups with VCC, Replogle/K562, Nadig/HepG2 and Jiang/IFNB/K562 as four source-defined laboratories. Each arm has four laboratories and 60 training interventions, with the same target anchor. The external-source arm scores 28.82 Top-1 versus 30.70 for the same-source arm. Underlying cell counts and gene panels remain different and are documented. DTI likewise reports each auxiliary source: Davis/Human improve the fixed target model, while BioSNAP and the combined-source setting do not.

These experiments test real differences between released studies, although execution is simulated on one host. We do not present heterogeneous data as automatically beneficial. Appendix E also reports the completed three-seed direct/history controls within the source-defined protein and cell settings, including their negative results.

3. Why direct search can be worse than the fixed model

“Direct” denotes feedback-free LLM configuration search, not training one model once. Every method ultimately deploys one selected predictor; ten laboratories are data holders contributing to fitting, not a ten-model ensemble. Both direct and history search retain the model with the best development score, whereas “fixed” is a single predefined configuration. Retaining a better development model therefore need not retain a better test model.

We investigated this failure rather than changing the reported test selection. Reconstructed VCC/Tahoe predictions show that equal-laboratory and pooled-intervention macro aggregation agree exactly. Small gains on 20 VCC or 60 Tahoe development interventions can accompany much larger test decreases. Re-evaluation under 20 new paired development-option rosters further shows that Qwen/VCC and Tahoe's original development advantages are not stable to distractor/option-order changes. No models were retrained or reselected in this diagnostic.

4. Scope and presentation of the contribution

The original access comparison now has an explicit Labs column and is no longer used to attribute the benefit of additional data to the LLM loop. The same-access search table addresses that attribution directly. We describe the method as a common biological model-development interface with four editable configuration factors, distinct from open-ended model invention. Related work now discusses Virtual Biotech, AutoScientists and Helmsman. A separate 360-fit allocation comparison adds ordinary score-only and random promotion: it reproduces the original benefit over uniform allocation but shows that much of it is also attained by simpler policies. The revised conclusion states these results separately from collaborative fitting and proposal history.